Testing UFW Firewall Rules in Kali Linux Using VirtualBox NAT

1. Objective

The objective of this test is to verify whether the UFW (Uncomplicated Firewall) rules configured in Kali Linux correctly control incoming network traffic.

2. Environment

- **Host operating system:** Windows
- **Virtualization software:** Oracle VirtualBox
- **Guest operating system:** Kali Linux
- **Network mode:** NAT
- **Firewall:** UFW
- **Testing tools:** Python HTTP server, Windows PowerShell, `curl`, and UFW logs

3. Configured Firewall Rules

No.	Port / Source	Action	Purpose
1	TCP port 22	ALLOW IN	Permit incoming SSH connections
2	TCP port 80	DENY IN	Block incoming HTTP connections
3	Port 443	ALLOW IN	Permit matching traffic on port 443
4	`198.51.100.0/24`	DENY IN	Block incoming traffic from the specified source range

5	TCP port 22 (IPv6)	ALLOW IN	Permit incoming IPv6 SSH connections
6	TCP port 80 (IPv6)	DENY IN	Block incoming IPv6 HTTP connections
7	Port 443 (IPv6)	ALLOW IN	Permit matching IPv6 traffic on port 443

4. Testing Procedure

Step 1: Verify the firewall status

In Kali Linux, execute:

sudo ufw status numbered

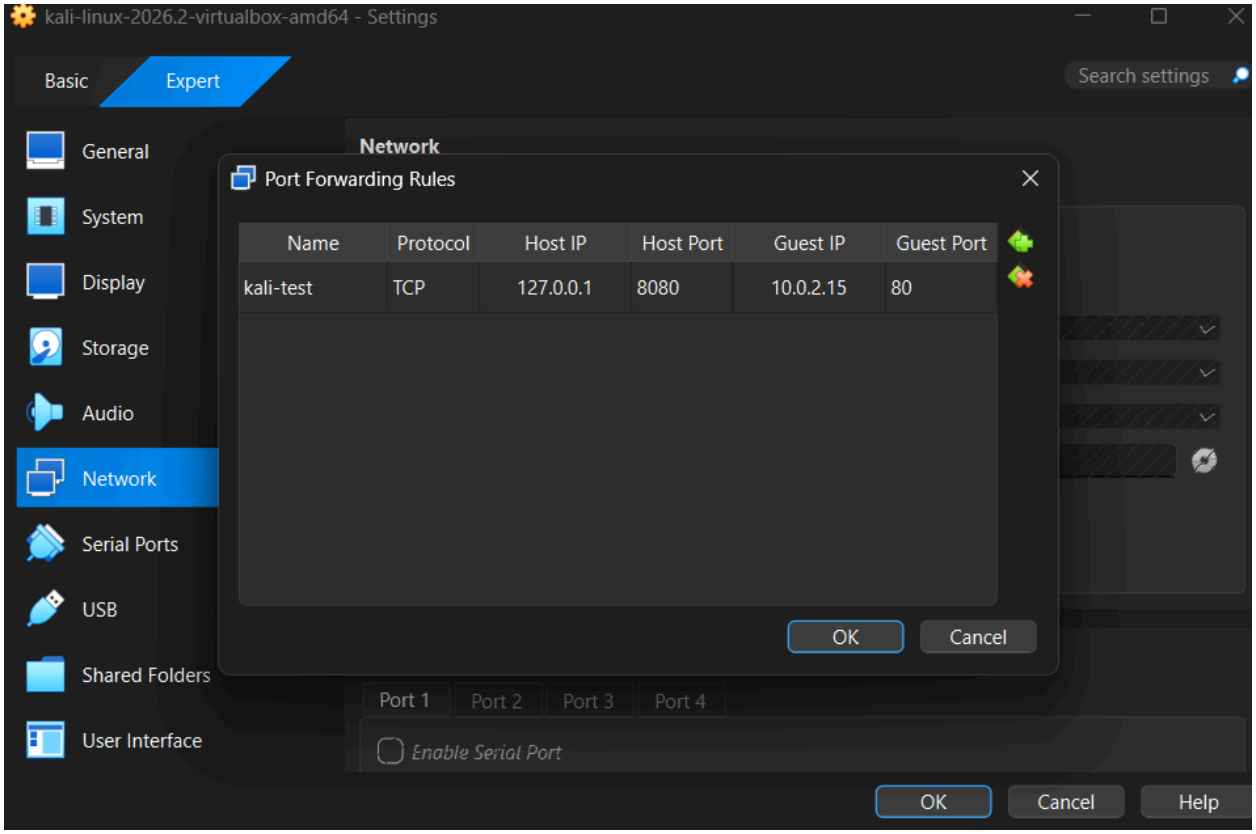
Expected result: UFW is active, and the configured firewall rules are displayed.

Step 2: Configure VirtualBox port forwarding

Because Kali uses NAT, configure a port-forwarding rule in VirtualBox:

- Protocol: TCP
- Host IP: `127.0.0.1`
- Host port: `8080`hawlet
- Guest IP: `10.0.2.15`
- Guest port: `80`

This forwards connections from Windows port 8080 to Kali port 80.

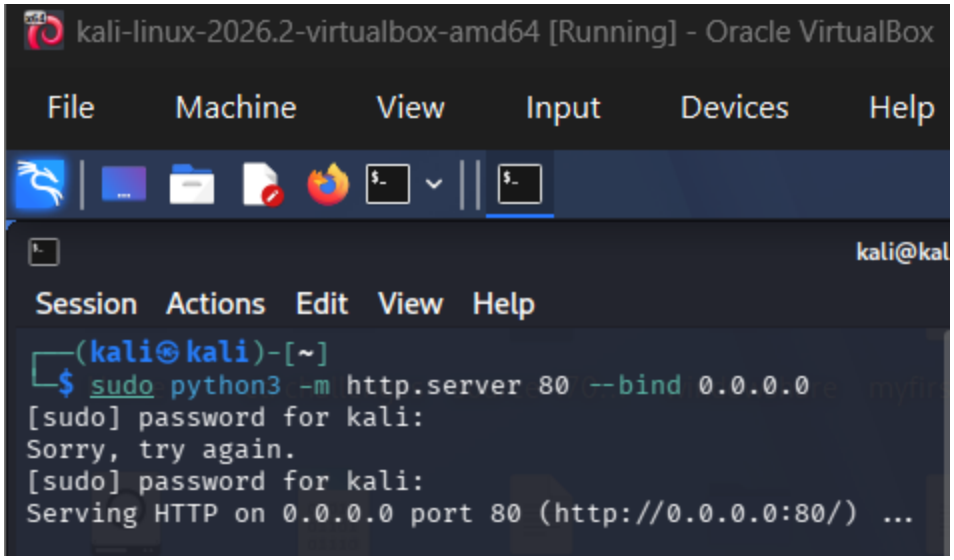


Step 3: Start a test HTTP server

In Kali Linux, execute:

sudo python3 -m http.server 80 --bind 0.0.0.0

The server listens for incoming HTTP connections on port 80.



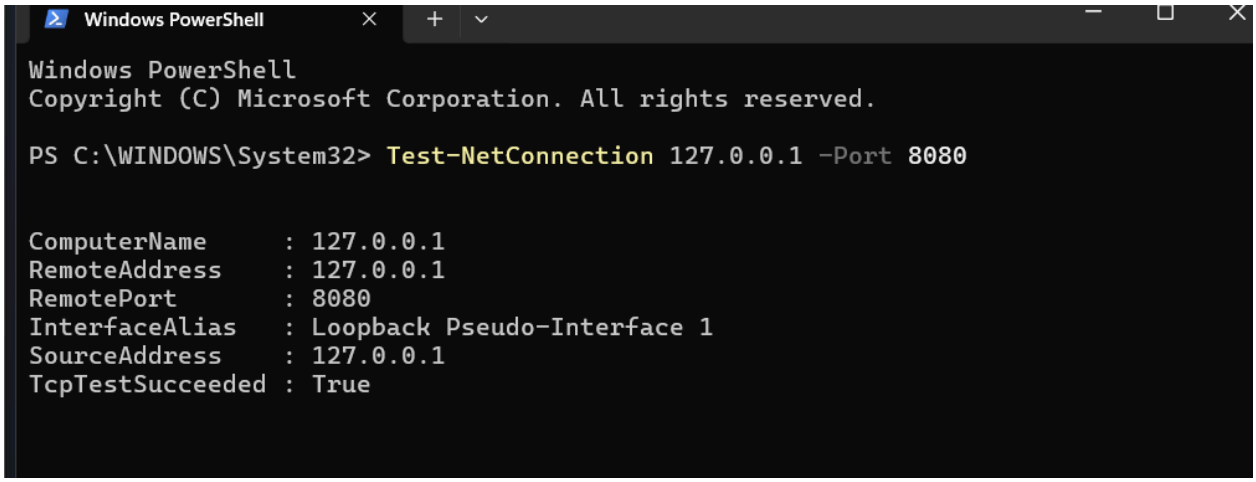
Step 4: Test connectivity from Windows

Open PowerShell and execute:

Test-NetConnection 127.0.0.1 -Port 8080

Observed result: `TcpTestSucceeded: True`.

This confirms that Windows can establish a TCP connection to the forwarded host port. It does not, by itself, establish whether Kali's firewall permits the connection.



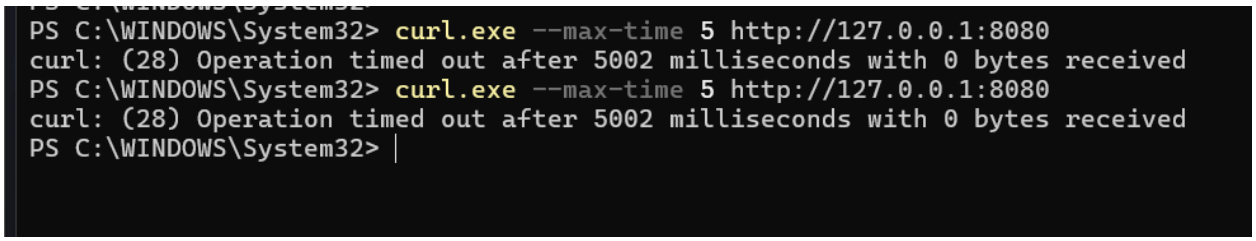
Step 5: Test the HTTP request

In Windows PowerShell, execute:

curl.exe --max-time 5 http://127.0.0.1:8080

Observed result: The request timed out after five seconds with zero bytes received.

This indicates that the HTTP request did not receive a response within the specified time. The timeout is consistent with traffic being blocked, but other network or server issues may cause the same result.



Step 6: Inspect UFW logs

In Kali Linux, enable firewall logging and inspect recent kernel logs:

sudo ufw logging medium

sudo journalctl -k --since "5 minutes ago" | grep UFW

The logs showed TCP SYN packets from `10.0.2.2` to `10.0.2.15` with destination port 80. These entries confirm that connection attempts reached the Kali network interface and were logged by UFW.

The observed entries were labelled `UFW AUDIT`, which does not independently prove that the packets were blocked.

5. Results and Analysis

Test	Result	Interpretation
UFW status	Active	Firewall is enabled
Firewall rule inspection	Port 80 DENY IN present	The deny rule is configured

Windows TCP connectivity test	Successful	The forwarded TCP port was reachable
HTTP request using curl	Timed out	No HTTP response was received within five seconds
UFW log inspection	TCP SYN packets recorded	Incoming connection attempts were logged

6. Conclusion

The UFW firewall was active, and the intended deny rule for TCP port 80 was configured. Testing from Windows through VirtualBox NAT showed that TCP connection attempts reached Kali, while the HTTP request timed out.

However, the available results do not conclusively prove that UFW blocked the traffic. A log entry explicitly identifying a blocked packet, such as `UFW BLOCK` with destination port 80, or equivalent verified firewall-rule evidence, is needed to confirm that conclusion.

7. Recommendation

Repeat the test while monitoring UFW logs and confirming the packet-handling rules. Record the results of both an allowed-port test and a denied-port test to demonstrate that the firewall correctly enforces its configured policy.